ATLAS UNIVERSITY ISTANBUL

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

Project

STUDENT EXAMINATION SCORE PREDICTION USING LINEAR REGRESSION

A Comparative Study of Two Linear Regression Models

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ABSTRACT

This study investigates the application of linear regression to predict students' examination scores using the Student Performance Factors dataset. The main objective was to compare two models: Model A, which uses Attendance and Hours_Studied, and Model B, which uses all 19 independent variables.

The dataset contains 6,607 records and 20 columns, including 19 independent variables and the target variable, Exam_Score. Data preparation included missing-value investigation, exploratory data analysis, numerical imputation, categorical imputation and one-hot encoding. The data was divided into training and testing subsets using an 80:20 split.

Both models were trained and evaluated using R-squared (R^2), Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). Model B achieved an R^2 of 0.7696, compared with 0.5890 for Model A, and recorded lower error values on the test set. The results indicate that Model B provided better predictive performance in this experiment.

CHAPTER ONE: INTRODUCTION

1.1 Background

Student academic performance may be associated with several factors, including study habits, attendance, previous academic performance, access to learning resources and other educational circumstances. Predicting examination scores is one way to investigate how information about these factors can be used to estimate academic outcomes.

Linear Regression is a supervised learning algorithm used to predict a continuous numerical outcome by modelling the relationship between a dependent variable and one or more independent variables. This project applies linear regression to predict students' examination scores.

1.2 Research Problem

A prediction model that uses only a small number of student-related variables may omit other useful information. This study therefore compares a model based on two selected variables with a model based on all 19 independent variables in the dataset.

1.3 Research Question

How accurately can linear regression predict students' examination scores using Attendance and Hours_Studied compared with using all 19 independent variables?

1.4 Aim and Objectives

The aim is to develop and compare two linear regression models for predicting students' examination scores. The objectives are to:

- Inspect and describe the Student Performance Factors dataset.
- Identify missing values, duplicate records and unusual observations.
- Perform exploratory data analysis.

- Prepare numerical and categorical variables for regression.
- Develop Model A using Attendance and Hours_Studied.
- Develop Model B using all 19 independent variables.
- Evaluate and compare the models using standard regression metrics.
- Select the model with the stronger test-set performance.

CHAPTER TWO: DATASET DESCRIPTION

2.1 Dataset Source

The study used the Student Performance Factors dataset obtained from Kaggle. The dataset contains student-related information and an examination-score outcome.

2.2 Dataset Structure

Description	Value
Total records	6,607
Total columns	20
Independent variables	19
Dependent variable	1
Numerical independent variables	6
Categorical independent variables	13

2.3 Target and Predictor Variables

The dependent variable is Exam_Score, which the models attempt to predict. The six numerical independent variables are Hours_Studied, Attendance, Sleep_Hours, Previous_Scores, Tutoring_Sessions and Physical_Activity.

The 13 categorical independent variables are Parental_Involvement, Access_to_Resources, Extracurricular_Activities, Motivation_Level, Internet_Access, Family_Income, Teacher_Quality, School_Type, Peer_Influence, Learning_Disabilities, Parental_Education_Level, Distance_from_Home and Gender.

CHAPTER THREE: DATA PREPROCESSING AND EXPLORATION

3.1 Data Inspection and Missing Values

Initial inspection identified 235 missing values across three categorical variables. Teacher_Quality contained 78 missing values, Parental_Education_Level contained 90, and Distance_from_Home contained 67. No duplicate records were identified.

Variable	Missing values
Teacher_Quality	78
Parental_Education_Level	90
Distance_from_Home	67
Total	235

Missing numerical values were prepared for median imputation, while missing categorical values were prepared for most-frequent-category imputation within the preprocessing pipelines.

3.2 Outlier Investigation

The Interquartile Range (IQR) method was applied to Exam_Score. The first quartile was 65, the third quartile was 69, and the IQR was 4. This produced a lower boundary of 59 and an upper boundary of 75. The method flagged 104 potential outliers.

One observation at row index 1525 had an Exam_Score of 101, with Hours_Studied equal to 27 and Attendance equal to 98. The assignment instructions and available dataset information did not specify that the maximum possible score was 100. Therefore, the observation was retained rather than being labelled an error without evidence. All 6,607 records were retained for modelling.

3.3 Exploratory Data Analysis

Exploratory data analysis included the distribution of Exam_Score, a boxplot, a correlation heatmap for numerical variables and comparisons of examination-score averages across categorical groups.

Numerical variable	Correlation with Exam_Score
Attendance	0.5811
Hours_Studied	0.4455
Previous_Scores	0.1751
Tutoring_Sessions	0.1565
Physical_Activity	0.0278
Sleep_Hours	-0.0170

Attendance and Hours_Studied showed the strongest positive numerical correlations with Exam_Score among the six numerical predictors. These results informed the selection of Model A predictors. Correlation alone does not establish causation or guarantee predictive performance.

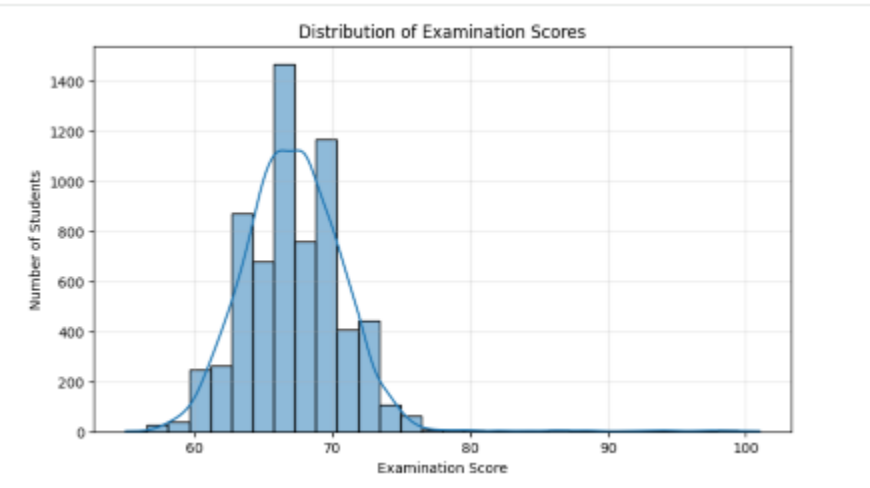


Figure 1. the Exam Score distribution histogram.

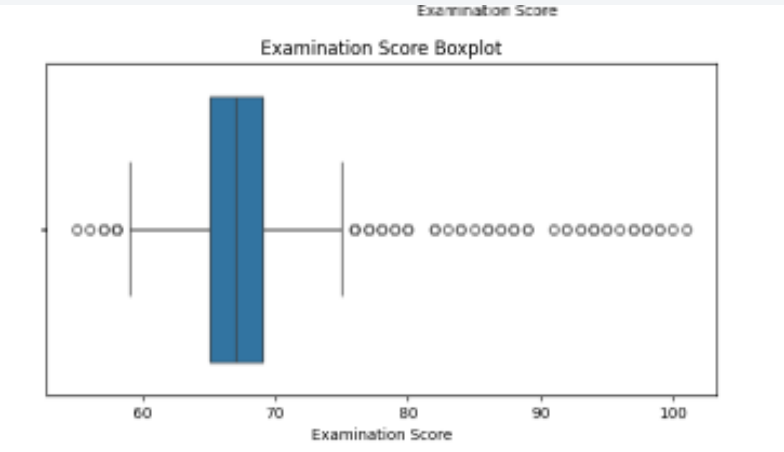


Figure 2. the Exam_Score boxplot.

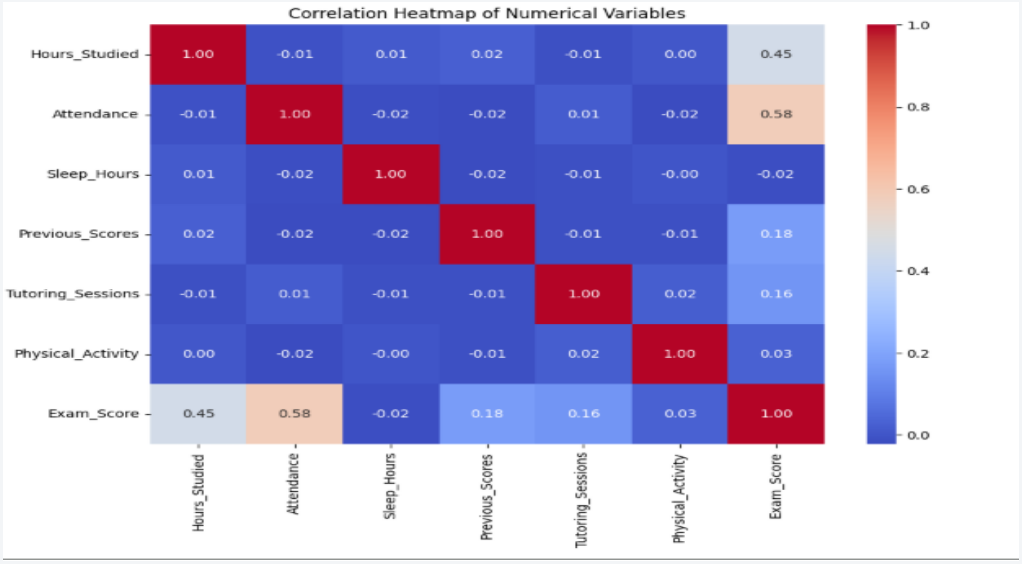


Figure 3.the numerical-variable correlation heatmap.

CHAPTER FOUR: RESEARCH METHODOLOGY

4.1 Research Approach

A quantitative experimental approach was used. Two linear regression models were trained on the same training subset and evaluated on the same held-out test subset to support a fair comparison.

4.2 Train-Test Split

The dataset was divided into 80% training data and 20% testing data using train_test_split with random_state=42. This produced 5,285 training records and 1,322 testing records. The same split was used for both models.

4.3 Preprocessing

Scikit-learn pipelines were used to perform preprocessing and regression. Numerical missing values were handled using median imputation. Categorical missing values were handled using most-frequent imputation, followed by One-Hot Encoding. The pipelines were fitted as part of model training, using training data only, to reduce the risk of information leakage from the test set.

4.4 General Linear Regression Equation

$$\hat{y} = b_0 + b_1X_1 + b_2X_2 + \ldots + b_nX_n$$

Here, $\hat{y}$ is the predicted examination score; b_0 is the intercept; b_1 through b_n are estimated regression coefficients; X_1 through X_n are the predictor variables; and n is the number of predictors in the stated

model. For Model B, categorical variables are transformed into indicator columns, so the encoded design matrix can contain more than 19 columns.

4.5 Model A Equation

$$\hat{y} = b_0 + b_1(\text{Attendance}) + b_2(\text{Hours_Studied})$$

Model A uses two predictors: Attendance and Hours_Studied.

4.6 Model B Equation

$$\hat{y} = b_0 + b_1X_1 + b_2X_2 + \ldots + b_{19}X_{19}$$

Model B uses all 19 original independent variables. The categorical variables are converted into numerical indicator columns using One-Hot Encoding before the regression estimator is fitted.

4.7 Software and Libraries

The experiment was conducted in Google Colab using Python, Pandas, NumPy, Matplotlib and Scikit-learn. The LinearRegression estimator was used within Scikit-learn pipelines.

CHAPTER FIVE: MODEL DEVELOPMENT

5.1 Model A

Model A was constructed using Attendance and Hours_Studied. The model pipeline applied numerical preprocessing and then fitted a Linear Regression estimator. It was trained on 5,285 records and used to predict the 1,322 test records.

5.2 Model B

Model B was constructed using all 19 independent variables. Its preprocessing pipeline handled the six numerical and 13 categorical variables before fitting the Linear Regression estimator. It used the same training and testing records as Model A.

5.3 Training Procedure

Both pipelines were successfully trained using the training subset. The test subset was reserved for evaluation and was not used to fit the models.

CHAPTER SIX: MODEL EVALUATION AND RESULTS

6.1 Evaluation Metrics

Four metrics were used to compare predictive performance. R^2 measures the proportion of variation in the target variable explained by the model relative to a baseline that predicts the test-set mean. Higher R^2 is generally better.

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$$

Mean Absolute Error measures the average absolute difference between actual and predicted scores. Lower MAE is better.

$$\mathbf{MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i|}$$

Mean Squared Error is the average squared prediction error. It penalizes larger errors more strongly. Lower MSE is better.

$$\mathbf{MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2}$$

Root Mean Squared Error is the square root of MSE and is expressed in the same units as the target score. Lower RMSE is better.

$$\mathbf{RMSE = \sqrt{MSE}}$$

In these expressions, y_i is the actual score, $\hat{y}_i$ is the predicted score, $\bar{y}$ is the mean actual score, and n is the number of test observations.

6.2 Final Evaluation Results

Evaluation metric	Model A (2 predictors)	Model B (19 predictors)
R-squared (R^2)	0.5890	0.7696
MAE	1.4690	0.4524
MSE	5.8091	3.2560
RMSE	2.4102	1.8044

Model B achieved a higher R^2 than Model A. It also recorded lower MAE, MSE and RMSE values. Thus, Model B produced better predictive performance on the held-out test dataset.

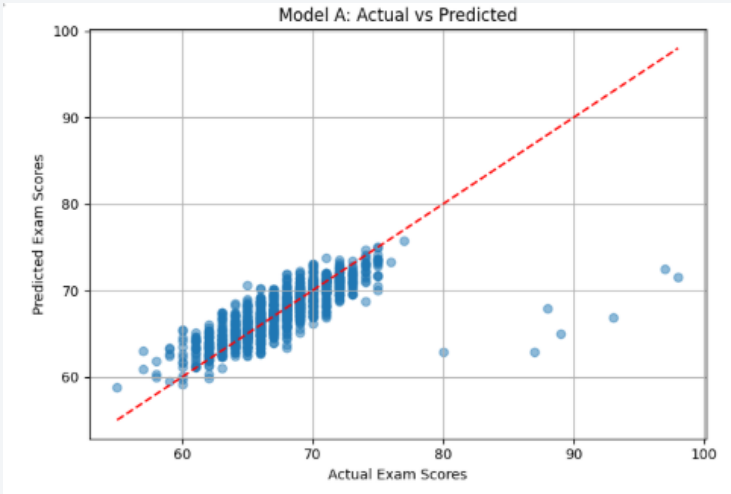


Figure 4. the Model A actual-versus-predicted scatterplot.

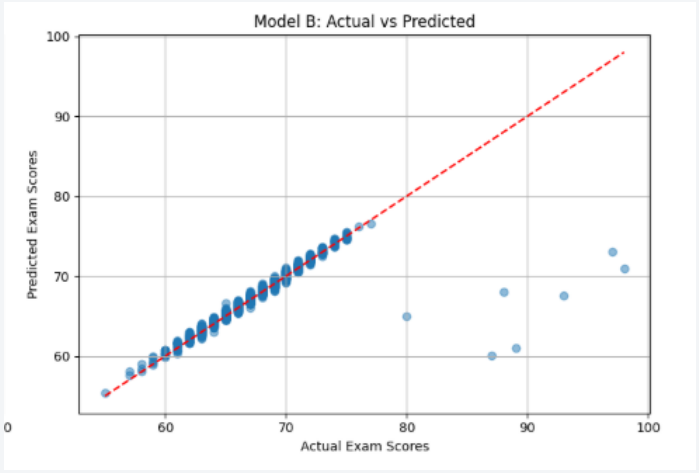


Figure 5.the Model B actual-versus-predicted scatterplot.

6.3 Residual Analysis

A residual is the difference between an actual value and its predicted value. Residuals were calculated as follows:

$$e_i = y_i - \hat{y}_i$$

The mean residual for Model A was −0.0123, while the mean residual for Model B was −0.0492. Both values are close to zero, indicating little average signed error on the test set. The negative values indicate a very small average tendency to overpredict. Mean residual alone does not establish that all regression assumptions are satisfied.

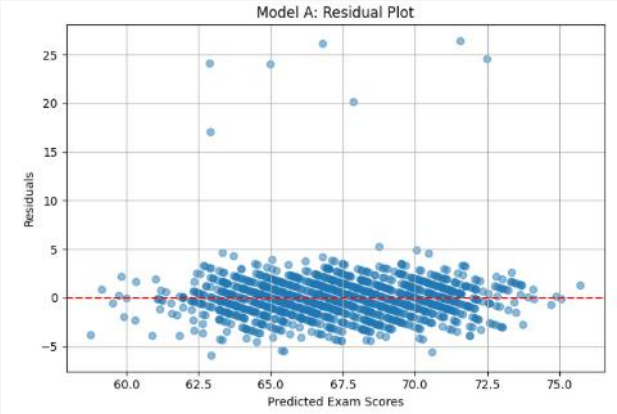


Figure 6. the Model A residual plot.

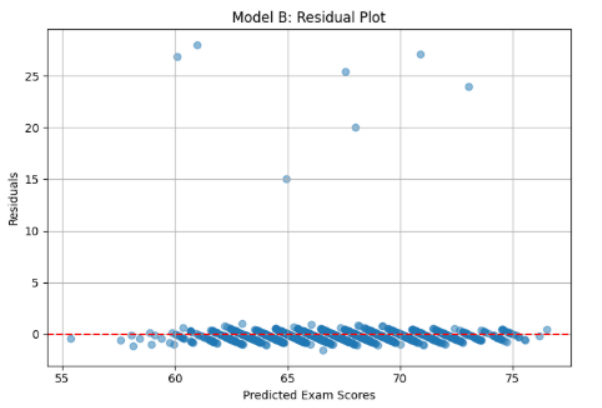


Figure 7. the Model B residual plot.

CHAPTER SEVEN: DISCUSSION

The results show that Model B performed better than Model A across all four evaluation metrics. Model A achieved an R^2 of 0.5890, indicating that it explained approximately 58.9% of the variation in examination scores on the test set. Model B achieved an R^2 of 0.7696, corresponding to approximately 77.0%.

Model B's MAE of 0.4524 was lower than Model A's MAE of 1.4690. Its MSE and RMSE were also lower. The actual-versus-predicted plots were visually consistent with this result: Model B's predictions appeared more closely aligned with the perfect-prediction reference line.

The comparison suggests that the additional student-related variables supplied useful predictive information in this dataset. It does not demonstrate that those variables cause changes in examination scores. The result is a predictive comparison on one held-out split, not proof that Model B will perform equally well on every new population or dataset.

CHAPTER EIGHT: LIMITATIONS

- The experiment used one dataset, so results may not generalize to all student populations.
- Linear regression models linear relationships in the encoded predictors and may not capture more complex nonlinear patterns.
- One-Hot Encoding can create many model coefficients, making interpretation more complex.
- Missing values required imputation, and the chosen imputation strategies may affect results.
- The score of 101 was retained because the available instructions did not establish a maximum score.
- The comparison used one 80:20 train-test split. Repeated cross-validation could provide a more robust estimate of generalization performance.
- The study evaluates prediction and does not establish causal relationships.

CHAPTER NINE: CONCLUSION

This study developed and compared two linear regression models for predicting students' examination scores using the Student Performance Factors dataset. Model A used Attendance and Hours_Studied, while Model B used all 19 independent variables.

Both models were trained using 5,285 records and evaluated using 1,322 held-out records. Model A achieved an R^2 of 0.5890, MAE of 1.4690, MSE of 5.8091 and RMSE of 2.4102. Model B achieved an R^2 of 0.7696, MAE of 0.4524, MSE of 3.2560 and RMSE of 1.8044.

Model B therefore provided the better predictive performance in this experiment. The research question was answered by showing that the model using all 19 independent variables predicted examination scores more accurately than the model using only Attendance and Hours_Studied on the test dataset.

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Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., VanderPlas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

APPENDIX: EXPERIMENT CONFIGURATION

Setting	Value
Dataset	Student Performance Factors
Total records	6,607
Target variable	Exam_Score
Model A predictors	2
Model B predictors	19
Algorithm	Linear Regression
Training proportion	80%
Testing proportion	20%
Random state	42
Training records	5,285
Testing records	1,322
Numerical imputation	Median
Categorical imputation	Most frequent
Categorical encoding	One-Hot Encoding
Best-performing model	Model B